

BOBO-DIOULASSO, Burkina Faso

WMO: 655100

Lat: 11.160N

Lon: 4.331W

Elev: 461

StdP: 95.91

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.2	19.3	-7.4	2.1	28.6	-5.7	2.5	28.2	8.1	24.2	7.1	24.8	3.0	20	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.3	38.2	20.4	37.7	20.4	36.9	20.6	26.1	32.1	25.7	31.6	25.4	31.2	3.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.7	28.6	24.2	20.2	28.2	23.9	19.9	28.0	83.7	32.5	81.9	32.1	80.4	31.4	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.3	6.5	5.9	DB	15.2	40.1	1.4	0.7	14.1	40.6	13.3	41.0	12.5	41.5	11.4	42.0	
			WB	7.3	27.4	1.1	0.8	6.5	28.0	5.9	28.5	5.2	28.9	4.4	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	26.2	29.0	31.3	31.1	29.8	27.7	26.2	25.6	26.2	27.7	28.3	26.4
	DBStd	2.51	2.00	2.24	1.68	1.83	1.85	1.70	1.42	1.18	1.36	1.35	1.05	1.65
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6543	501	531	659	632	614	532	501	483	484	548	549	508
	CDD18.3	3502	243	298	401	382	356	282	242	225	234	289	299	250
	CDH23.3	40473	2624	3897	5897	5424	4672	3083	2059	1564	1898	3051	3581	2723
	CDH26.7	20151	1244	2156	3563	3181	2478	1302	656	403	642	1367	1855	1303
(14) Wind	WSAvg	3.1	3.5	3.4	3.1	3.4	3.6	3.5	3.1	2.9	2.4	2.4	2.4	3.2
(15) Precipitation	PrecAvg	1033	1	2	17	46	93	133	210	272	186	64	8	1
	PrecMax	1406	20	36	95	160	250	258	437	460	374	145	77	20
	PrecMin	626	0	0	0	0	21	38	68	99	75	2	0	0
	PrecStd	179	4	7	21	35	48	47	75	83	64	38	16	4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	38.4	39.2	39.2	38.2	36.0	33.2	31.8	33.2	35.2	36.0	35.6
		MCWB	16.9	18.8	19.5	21.1	23.3	24.9	24.8	25.0	25.1	22.4	19.8	18.0
	2%	DB	34.8	37.2	38.3	38.2	37.0	34.2	32.0	30.8	32.1	34.2	35.1	34.5
		MCWB	16.3	18.5	19.4	21.4	24.1	24.9	24.7	24.7	24.9	23.2	19.7	17.2
	5%	DB	33.6	36.3	37.8	37.4	35.9	33.1	31.0	29.9	31.2	33.3	34.6	33.7
		MCWB	16.0	18.0	19.3	21.9	24.3	24.8	24.5	24.6	24.8	23.8	19.7	16.9
	10%	DB	32.4	35.2	36.9	36.4	34.8	32.1	30.1	29.0	30.2	32.5	33.9	32.7
		MCWB	15.6	17.4	19.4	22.4	24.4	24.5	24.3	24.4	24.6	24.1	19.7	16.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	23.6	25.0	26.0	27.0	26.2	25.7	25.7	26.1	26.1	24.7	22.8
		MCDB	30.2	33.8	33.4	34.5	33.3	32.3	30.6	29.6	31.1	31.3	31.3	30.2
	2%	WB	18.5	22.4	24.2	25.2	26.2	25.7	25.2	25.2	25.5	25.5	24.0	20.5
		MCDB	31.1	32.2	33.2	33.5	32.8	31.7	30.0	29.2	30.4	31.0	30.7	29.2
	5%	WB	17.2	21.4	23.6	24.8	25.7	25.3	24.9	24.9	25.1	25.1	23.3	18.6
		MCDB	31.2	31.2	32.5	33.0	32.3	31.2	29.6	28.8	29.9	30.7	29.9	31.2
	10%	WB	16.3	20.1	22.9	24.4	25.3	25.0	24.6	24.6	24.8	24.7	22.7	17.6
		MCDB	30.8	31.6	31.7	32.4	31.9	30.7	29.0	28.2	29.2	30.2	29.2	30.7
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	12.1	11.3	10.6	9.8	8.7	7.6	7.0	8.1	9.8	12.1	12.8
		MCDBR	13.5	12.7	12.2	11.6	11.0	10.0	8.9	8.2	9.5	10.9	12.7	13.5
		MCWBR	5.6	5.5	5.2	4.3	3.1	3.4	3.1	3.3	3.7	3.7	4.8	5.3
	5% WB	MCDBR	12.4	10.7	10.5	10.4	9.7	9.1	8.2	7.7	8.7	9.9	11.0	12.8
		MCWBR	5.8	6.1	4.7	3.6	3.7	3.6	3.3	3.3	3.8	3.9	4.3	5.8
(40) Clear-Sky Solar Irradiance	taub	0.546	0.615	0.736	0.670	0.614	0.570	0.511	0.488	0.493	0.547	0.523	0.500	
	taud	1.848	1.709	1.524	1.715	1.804	1.924	2.128	2.205	2.164	1.982	1.950	1.972	
	Ebn at Noon	763	727	653	693	718	741	790	821	823	773	777	792	
	Edh at Noon	205	243	297	244	218	191	157	147	154	182	183	177	
(44) All-Sky Solar Radiation	RadAvg	5.68	5.90	6.08	6.04	6.00	5.61	5.23	4.98	5.32	5.81	5.62	5.51	
(45)	RadStd	0.29	0.29	0.28	0.24	0.19	0.21	0.16	0.20	0.21	0.15	0.17	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page